

Stochastics

(4. SDEs and Ito Integrals)

$$\text{ODEs} \rightarrow \frac{dy}{dt}, \frac{dy}{dx} \quad \} \quad F = \frac{dp}{dt} \leftarrow$$

$$\text{PDEs} \rightarrow \frac{\partial y}{\partial x} \quad \} \quad \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} \quad (1\text{-D heat eq.})$$

$$\text{SDEs} \rightarrow \textcircled{dw}$$

Black-Scholes
(PDE)

$$\frac{dx}{dt}, \textcircled{\frac{dw}{dt}}$$

$$dx = (\quad) dt + \underline{\underline{(\quad) dw}}$$

Stochastic Differential Equation

- A one-dimensional stochastic differential equation (SDE) can be described as

x_t $x_t + dx_t$
 t $t+dt$

$$dX_t = \underbrace{\mu(X_t, t)dt}_{\text{(drift)}} + \underbrace{\sigma(X_t, t)dW_t}_{\text{(volatility) = random}}$$

$\mathbb{E}[dx] = 0$

We can model asset prices using the above form. Note that the first term is a deterministic trend or drift component whereas the second term is the random or stochastic or diffusion component. Both drift and volatility can be function of time (t) and level (X_t) in general

The process X_t in integral form is given as

$\mathbb{E}[x_t | \mathcal{F}_u] = x_u$
 $\mathbb{E}[\Delta x_t | \mathcal{F}_u] = 0$ *Solution*

$t = 1 \text{ year}$

$$X_t = X_0 + \int_0^t \mu(X_s, s) ds + \int_0^t \sigma(X_s, s) dW_s$$

Mean of X_t

- If we look at the integral form, notice that the last term is an Ito integral which is a martingale with mean = 0. The mean of X_t is

$$\rightarrow X_t = \underline{X_0} + \int_0^t \underline{\mu(X_s, s)} ds + \underbrace{\int_0^t \sigma(X_s, s) dW_s}_{\text{(Ito Integral)}}$$

$$\mathbb{E}[X_t] = \underline{X_0} + \mathbb{E} \left[\int_0^t \mu(X_s, s) ds \right]$$

Condition for Martingale

► Remember that if X_t is a martingale, then X_t must satisfy

$$\mathbb{E}[X_t | \mathcal{F}_s] = X_s \quad \checkmark$$

Clearly this will be satisfied if there is no drift. i.e.,

$$\underline{\mu(X_t, t)} = 0 \quad \checkmark$$

Martingales have SDEs with no drift i.e.,

$$dX_t = \underline{\underline{\sigma(X_t, t)dW_t}}$$

(0).dt
X

Ito's Lemma

- If X_t follows drift-diffusion process i.e., $dX_t = \mu(X_t, t)dt + \sigma(X_t, t)dW_t$ and $f(t, X_t)$ is a twice-differentiable function, then the SDE of f is given as

$$df = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dx + \frac{1}{2} \sigma^2 \frac{\partial^2 f}{\partial x^2} dt$$

We can use Ito's lemma or Taylor series to derive SDE for any process. We can use Ito integral to arrive at the process from it's SDE. We need to use Taylor series expansion till 2nd order with the following rules in mind

$$\Delta f = \frac{\partial f}{\partial t} \Delta t + \frac{\partial f}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} (\Delta x)^2$$

$O(\Delta t)$

(Ito multiplication table)

$\times$	dt	dW_t
dt	$dt^2 = 0$	$dt dW_t = 0$
dW_t	$dW_t dt = 0$	$dW_t^2 = dt$

$\rightarrow dt \cdot 2 \cdot \sqrt{dt}$

Case I : $F = F(\underline{W_t})$

► For functions with explicit time dependence on only W_t , the SDE is given as

$$\rightarrow \underline{dF} = \frac{dF}{\underline{dW_t}} dW_t + \frac{1}{2} \frac{d^2 F}{\underline{dW_t^2}} dt$$

Exercise –

1) Find the SDE for

a) W_t^2 ✓

b) $\exp(W_t)$ ✓

2) Is $\underline{W_t^2}$ a martingale ?

Case II : $F = F(t, W_t)$

► For functions with explicit time dependence on both t and W_t , the SDE is given as

$$dF = \left(\frac{\partial F}{\partial t} + \frac{1}{2} \frac{\partial^2 F}{\partial W_t^2} \right) dt + \frac{\partial F}{\partial W_t} dW_t$$

Exercise –

1) Find the SDE for ✓

a) $W_t^2 - t$. (Is it a martingale ?)

b) $\exp(W_t)$

Case III : $F = \underline{\underline{F(S_t)}}$

► For a function $F(S_t)$ where S_t satisfies the SDE

$$\rightarrow \underline{\underline{dS_t}} = \underline{\underline{\mu S_t dt}} + \underline{\underline{\sigma S_t dW_t}} \quad , \quad dS_t = \mu(S_t) dt + \sigma(S_t) dW_t$$

$\mu(S_t) \rightarrow \mu S_t$ $\sigma(S_t) \rightarrow \sigma S_t$

$$dF = \left(\mu S_t \frac{dF}{dS_t} + \frac{1}{2} \sigma^2 S_t^2 \frac{d^2 F}{dS^2} \right) dt + \sigma S_t \frac{dF}{dS_t} dW_t$$

Exercise –

Let's say $dS_t = \mu S_t dt + \sigma S_t dW_t$.

- 1) Find the SDE for $F = \log S$
- 2) Find the integral form of the process F_t and S_t .
- 3) What is the name of this process the S_t follows ?
- 4) Comment on the distribution of F_t and S_t . Find $\mathbb{E}[F_t]$ and $\mathbb{E}[S_t]$

Case IV : $F = F(t, S_t)$

► For a function $F(t, S_t)$ where S_t satisfies the SDE

$$\underline{dS_t} = \underline{\mu S_t dt} + \underline{\sigma S_t dW_t}$$

$$\underline{dF} = \left(\frac{\partial F}{\partial t} + \mu S_t \frac{\partial F}{\partial S_t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 F}{\partial S^2} \right) dt + \sigma S_t \frac{\partial F}{\partial S_t} dW_t$$

Exercise –

Let's say $dS_t = \mu S_t dt + \sigma S_t dW_t$. Find the SDE for $F = \exp(\underline{tS_t})$

Case V : $F = F(\underline{S}_t^{(1)}, \underline{S}_t^{(2)})$

► For a function $F(S_t^{(1)}, S_t^{(2)})$ where $S_t^{(i)}$ satisfies the SDE

$$\rightarrow \underline{dS}_t^{(i)} = \underline{\mu_i S}_t^{(i)} dt + \underline{\sigma_i S}_t^{(i)} dW_t^{(i)} \quad \text{where } i = 1, 2$$

$$\text{and } \underline{dW_1} \cdot \underline{dW_2} = \underline{\rho dt}$$

[] (ρ) ←

proceed with bi-variate Taylor series expansion as usual. Replace $dW_1 \cdot dW_2 = \rho dt$. Here ρ is the correlation between the two Brownian motions. We face this differential in case of Basket securities.

Exercise –

Derive the SDE for F in the case above.

Ito Product Rule

► Let X_t and Y_t be two stochastic processes, where

$$\begin{aligned} \rightarrow \underline{dX_t} &= \underline{a(X_t, t)dt} + \underline{b(X_t, t)dW_t^{(1)}} \\ \rightarrow \underline{dY_t} &= \underline{c(Y_t, t)dt} + \underline{d(Y_t, t)dW_t^{(2)}} \end{aligned}$$

Define a function F as the product i.e., $F = \underline{X_t Y_t}$

$$\underline{\frac{d(xy)}{dt}} = \underline{x \frac{dy}{dt}} + \underline{y \frac{dx}{dt}}$$

Then $\underline{d(X_t Y_t)} = \underline{X_t dY_t} + \underline{Y_t dX_t} + \underline{dX_t dY_t}$

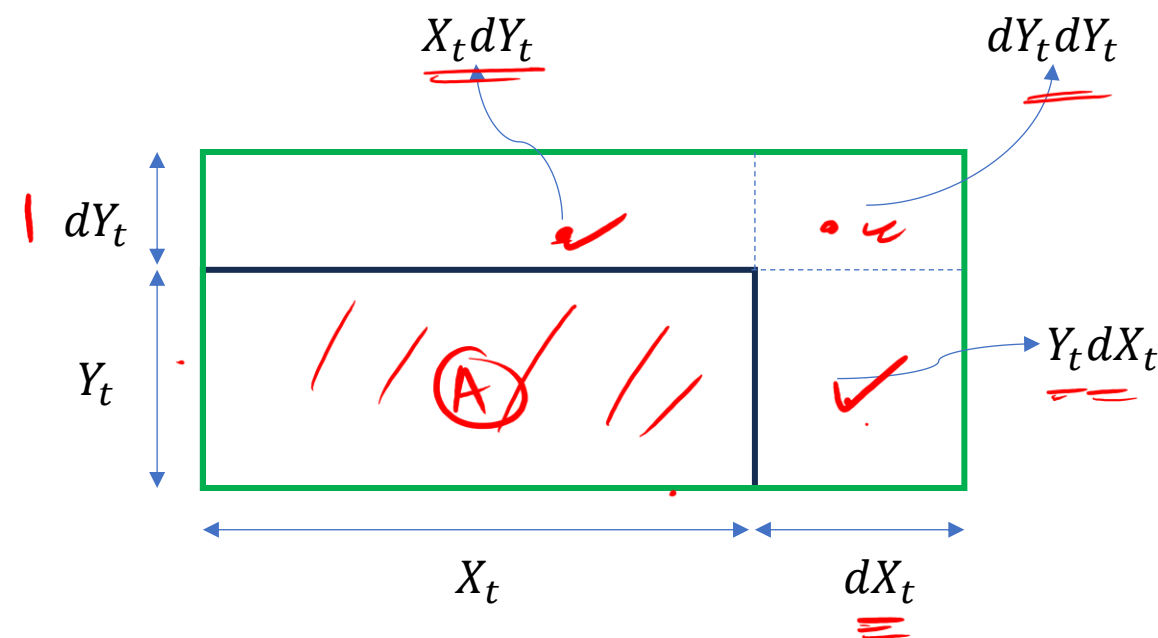
Note that in ordinary calculus, the third term does not appear i.e., $d(X_t Y_t) = X_t dY_t + Y_t dX_t$

Ito Product Rule (Geometric Interpretation)

- Note that X_t and Y_t can be represented as the sides of a rectangle and the product ($X_t Y_t$) represents the area of the rectangle. In ordinary calculus, the corner gets dropped, but in Ito product we cannot drop it because of quadratic variation.

$$A = \textcircled{xy}$$

$$dA = d(xy)$$



Ito Quotient Rule



► Let X_t and Y_t be two stochastic processes, where

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t^{(1)}$$

$$dY_t = c(Y_t, t)dt + d(Y_t, t)dW_t^{(2)}$$

Define a function F as the ratio i.e., $F = \frac{X_t}{Y_t} = \cancel{\times} \cdot \left(\frac{1}{\cancel{Y}}\right)$

$$\text{Then } d\left(\frac{X_t}{Y_t}\right) = \frac{X_t}{Y_t} \left(\frac{dX_t}{X_t} - \frac{dY_t}{Y_t} - \left(\frac{dX_t}{X_t}\right) \left(\frac{dY_t}{Y_t}\right) + \left(\frac{dY_t}{Y_t}\right)^2 \right)$$

Ito Integral Formula I



$$\int_0^t \frac{dF}{dW_s} dW_s = F(W_t) - F(W_0) - \frac{1}{2} \int_0^t \frac{d^2 F}{dW_s^2} ds$$

Example :

$$\int_0^t \underline{\underline{e^{W_s} dW_s}}$$

Ito Integral Formula II



$$\int_0^t \frac{\partial F}{\partial W_s} dW_s = F(t, W_t) - F(0, W_0) - \int_0^t \left(\frac{\partial F}{\partial s} + \frac{1}{2} \frac{\partial^2 F}{\partial W_s^2} \right) ds$$

Example :

$$\int_0^t \underline{\underline{(sW_s + e^{W_s})}} dW_s$$

Ornstein-Uhlenbeck Process

- The Ornstein-Uhlenbeck process is a mean reverting process given by

$$\underbrace{(\underline{OU})}_{\bar{r}, \bar{\theta}} \quad \underline{dX_t} = -\gamma \underline{X_t} dt + \sigma \underline{dW_t}$$

This process is of special importance as we will see later. It's a popular model for short rate which display mean reverting process (Vasicek Model). More later!

Exercise - Show that solution to the above process is given as

$$\rightarrow \underline{X_t} = \underline{X_0} e^{-\gamma t} + \sigma \int_0^t e^{-\gamma(t-s)} dW_s$$

Also prove that

$$\underline{X_t} \sim N \left(\underline{X_0} e^{-\gamma t}, \underline{\frac{\sigma^2}{2\gamma} (1 - e^{-2\gamma t})} \right)$$